

(Action)

AGR401: RESEARCH METHODOLOGY

Man has universally adopted three methods to comprehend the world around him, namely: experience, reasoning and research.

The concept of research

Research has been very difficult to define or explain by Scholars and Laymen together. Many like to associate with the word research because it is novel, but "when asked what research is?" it is given different interpretations. Many people see research in different forms based on the stage at which they have contact with it as follows:

1. Some group especially scholars see research as investigation to finding solution to a problem.
2. Some see research as a positive outcome of an activity that is new or strange such as "invention" in physics or any other type of discovery. In this case any research activity that has not produced a result or discovery is not viewed that a research has taken places.
3. Some see research as a product or change that can be utilized to improve existing situation, this is an observation phenomenon, it is when one could feel, see and use it that they believe research has taken place, for example, a new model of a car or a new beverage or processed food item such as powdered yam.
4. Some see research as a means of researching, its used as a means of achieving the objectives of a planned research.

In a broad sense, research can be viewed as a complex word that can be explained broadly and theoretically as involving a chain of terminologies.

Therefore, Research can be defined as the systematic **rigorous investigation of a situation or problem** in order to generate new knowledge or to validate existing knowledge.

What scientific research entails

- a. Theory or principle
- b. Process
- c. Product and
- d. Utilization

Each of these terminologies involve activities that culminate to research as explained below

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- a) Theory or principles in simple research work such as dissertation, thesis or proposal, it is usually observed that the first stage is to develop what is known and content outline of the research or research outline with about five chapters, with chapter one titled introduction. The introduction has some subheadings, as stated below:
 - i. Background of the study
 - ii. Statement of the problem,
 - iii. Purpose of the study
 - iv. Significance of the study
 - v. Research question and hypotheses
- b) Research process: this is a step by step technique of carrying out a research process which may involve comfortably designed activities, prominent in research process are:
 - i. Literature reviews; it serves as a transition between theory and practices in research, it helps to identify a suitable framework for a research and relevant supportive research information to be investigated.
 - ii. Methodology; it involves practical activities to be carried out to lend support to the theories or assumption in chapter one to give room for a convincing proof.
 - iii. Data analysis: this involves practical activities carried out in analyzing or breaking into understandable form, of the data collected making use of relevant tools.
- c) Product: this is the outcome of a research endeavor, it can either be in an observable forms, form of statements, confirmation of ideas/records of events, conclusions or improvement on event. They are most often related to theories in chapter one especially research question and hypotheses all claiming relevance to this solution of the problem of the research.
- d) Utilization: this is the most important aspect of research to a non-researcher. It involves the implementation of the findings of a research and its positive effect on development.

Research design

Research design is the plan structure and strategy of investigation conceived so as to attain answers to research questions and to control variables.

Research design is the framework or organizational approach to an investigation so conceived as to obtain appropriate solutions of the problem being investigated.

This plant is the overall scheme or programme for carrying out the research. This includes what the investigation will do, how and where he will do the investigation and the implications to the final analysis of the data collected.

Purpose of research design

There are three main purposes namely;

- a) To provide answer to questions
- b) To control variables and
- c) To outline conclusions

Design helps the researcher to obtain answers to the questions of research. It also helps to control the experimental, extraneous, and error variances of the particular research under study. Research design helps to outline clearly the possible conclusions; the researcher can reach by specifically outlining or suggesting the statistical test that can be made. It should be noted that research problems suggest research designs.

Types of research design

There are many types of research that a researcher can select from, depending on the direction and mission of the research. Two principal types of research designs useful in

Agricultural Technology include;

- a) Experimental research design
- b) Survey research design

Experimental research design

Experimental research design includes a study which makes use of experiments. An experiment is a carefully researcher-manipulated design of a study that make use of two or more groups involving treatment and control. The subjects for the study are randomly assigned to these two groups, with the groups undergoing pre-tests and post-tests treatments.

Survey research design

Survey research design is a research design that employs the study of large and small population by selecting and studying sample chosen from a population to discover the relative incidence, distribution and interrelations of sociological and psychological variables. This approach is normally referred to as sample survey, since by definition it links sample and population. Survey researchers are interested in the actual assessment of characteristics of the whole population of people. They would want to know, for example how many people

in a community adopted a particular innovation and the relationship between attitude toward new innovation and public support of the innovation.

Survey researchers, they study sample drawn from the population not entire population. The study of samples from which inferences about population can be drawn is needed because of the difficulties of studying the whole population.

Random sampling is almost always employed as it can greatly assist to furnish better information at lower cost, with greater efficiency and sometime, greater accuracy. Survey research can be conveniently classified by the following methods of obtaining information: personal interview, mail questionnaire and telephone interview.

Steps of scientific research method

The steps and sequence are as follows;

- 1- Identification and statement of the problem
- 2- Formulating a hypothesis
- 3- Planning the research design
- 4- Collecting the data
- 5- Organization and analysis of data
- 6- Interpreting result and drawing conclusions.
- 7- Reporting the findings

Characteristics of research process

The research process has five (5) main characteristics;

- 1- Research is systematic; there is rules/regulation for carrying out a research.
- 2- Research is logical: logic simply means the science of or method of reasoning. Or the ability to argue and convince.
- 3- Research is empirical: Research is real in that, it relies on observations and experiment, not merely on theory or speculation.
- 4- Research is deductive: by applying analytical procedures to the data collected, the researcher reduces the confusion of individual events and objects to more understandable categories of concepts.
- 5- Research is replicable and transmittable: consequently, individuals other than the researcher may use the results of a study and one researcher may build upon the research results of another.

Attributes of a good researcher

Enthusiasm: requires skill and concentration.

- ① Open-mindedness: requires that the researcher observes with a keen attentive, inquisitive and open mind.

- ② Common sense: much effort is lost or deviated when the researcher fails to use common sense for the answers to his/her research questions.

- ③ Role-taking ability: think of yourself as the user of the research, not just as the person who has generated it.

- ④ Inventiveness: the researcher must be able to develop good hypotheses, finding solution to problems of financial resources, responding to emergencies during the conduct of research, finding new ways to analyses data and coming up with convincing interpretation of results.

- ⑤ Confidence on one owns judgement: researcher must be able to follow his own glean and bent, however, inadequate they may be.

- ⑥ Consistency and care about details: there is no substitute for accuracy and the cares needed to keep complete records organize and State fact precisely and proof read carefully.

- ⑦ Ability to communicate: this is basic to the practice of good research particularly in writing the final report.

- ⑧ Honesty: research respect integrity and avoid dishonesty or sloppiness (fraud) e.g. rigged experiment or presentation of faked results.

- ⑨ Errors in research

Error in research means failure to collect the data accurately. The common errors manifest in the following ways:-

Concepts and hypothesis construction: construction of ambiguous concepts and poor operational definition of concept can lack face validity, lack of face validity leads to lack of reliability, since the latter is dependent upon the former.

Construction of research instrument (questionnaire): even if the concept is clear and well defined, worded items in the measuring instrument can cause lack of reliability if the respondents cannot understand them.

Sampling procedures: bad/incorrect sampling procedures can lead to unrepresentative data, resulting in an inability to generalize the finding beyond the specific sample of respondents used in the study.

Data gathering: many potential sources of error are found in the actual data gathering situation.

Coding: the person coding the data or transcribing/transferring them from the questionnaire to the computer or from tape recording to transcript page can make simple clerical errors such as writing an incorrect number.

Data analysis: this involves misinterpretation of data during the data analysis stage, one makes errors at this strange by drawing incorrect conclusions from the data, or by saying that data show something that they do not show, such often occurs when the data analysis is biased or not properly trained.

Research Hypothesis is the statement created by researchers when they speculate upon the outcome of a research or experiment. Or to say Hypothesis testing is the process by which the belief is tested by statistical means.

For instance a hypothesis, "female students are higher than male students". When we apply the student's data in to statistical means and based on the results obtained from the sample the hypothesis would either be accepted or rejected. The hypothesis to be tested is termed the Null-hypothesis, designated " H_0 ", while alternative hypothesis designated " H_a ".

There are four possible results when we test the hypothesis H_0 .

- i. we accept a true hypothesis - a correct decision
 - ii. we reject a false hypothesis - a correct decision
 - iii. we reject a true hypothesis - an incorrect decision. This is known as type 1 error.
 - iv. we accept a false hypothesis - an incorrect decision. This is known as type 2 error.
- Effort are made to avoid any type of error but as it is not possible to make correct decision with 100% certainty when a hypothesis is tested by sampling, there is always a possibility of either type 1 or type 2 error.

The errors are mutually exclusive, error can be type 1 or type 2 but not both.

The errors are split in to two type because there are situation where it is much more important to avoid one type of error rather than the other.

For example, if we have the hypothesis: "the water is shallow" the consequences of a types 2 error could be

Fatal for a non - swimmer. In social research or survey research, hypothesis could be derived from research questions, while in Experimental research hypothesis could be derived from Objectives.

Characteristics of Hypothesis

- i. A discriminating. E.g. significant difference or better than, greater than and so on.
- ii. Quantitative value. E.g. mean rating, mean score, mean height, mean weight etc.

- iii. Variables. Such as male, female, experience, non-experienced, men, women, girl, boys etc.
- iv. Direction of focus. This is the main theme or object and category of respondents e.g. the number or population is the theme, while Teachers and student is the category of respondent.

Chi-square (X^2) distribution it's an important method of hypothesis testing and is used when it is wished to compare an Actual or Observed distributions with a hypothesized expected distribution. It's also called goodness of fit test.

$$\text{Formula } X = \sum \frac{(O-E)^2}{E}$$

Where O=the observe frequency value

E=the expected frequency value

Exercise: If a farmer who reside in a village and extension agent in that community recommend the use of or application of 450kg of 4types of fertilizers of NPK, UREA, CAN and SSP in the ratio of 4:3:2:1 respectively. The farmer applied 190kg of NPK, 140kg of UREA, 70kg of CAN and 50kg of SSP. Did the farmer complied with extension agent directive?

Hypothesis: H_0 = the farmer did not adhere to agents recommendation.

H_a = the farmer did adhere to agent recommendation.

The first stage in the solution is to calculate the expected frequency(fe) for each category which will then be compared with actual frequency(fo) observe frequency as shown below.

$$4:3:2+1 = 10 \text{ ratio}$$

Fe-computation	Fe	Fo	Fo-Fe	(Fo-Fe) ²	(Fo-Fe) ² /Fe
NPK 4/10 x 450 = 180	180	190	10	100	0.56
UREA 3/10 x 450 = 135	135	140	5	25	0.19
CAN 2/10 x 450 = 90	90	70	-20	400	4.44
SSP 1/10 x 450 = 45	45	50	5	25	0.56
Total	450	450			5.75= X^2 -cal.

It's necessary to find the appropriate X^2 value from X^2 distribution table and this is done by establishing the degree of freedom. This is found by multiplying the number of rows in table less one, by the number of columns less one i.e. $df = (\text{Rows}-1)(\text{columns}-1)$

$$= (4-1)(2-1)$$

$$= (3)(1) = 3df$$

Therefore, the value cut-off point of χ^2 -distribution table for 3 degree of freedom at the 5% level is 7.815 or 7.82

Decision: If $\chi^2_{\text{cal}} > \chi^2_{\text{tab}}$ mean reject the H_0 and accept H_a .

If $\chi^2_{\text{cal}} < \chi^2_{\text{tab}}$ mean accept H_0 and reject H_a

Sampling techniques

Experience over the years has shown that sampling is a problem to most students especially at the National Diploma and Higher National Diploma. A sample is a smaller representation of the whole group that is being studied. As the researcher plans the research design, he/she must also start thinking about the appropriate sample or population of groups, individuals or behaviours upon which to test the hypothesis.

A population (or universe) is that body of individual in which the researcher is interested and to which he/she will generate his/her findings. Once a sample is carefully selected, according to well defined sampling techniques, reliable and valid inference can be drawn about the larger population.

Reasons for carrying out sampling

There are three main reasons why we sample;

1. To take a complete enumeration that describe entire universe (population): for example, a patient who wants to know his/her blood group should take a sample of the blood rather than take all the blood for testing to do the latter can destroy the patient.
2. We sample because of cost consideration/implication: sampling allows a researcher to obtain essentially a useful information/data, at a lower cost. Provided the sampling is done according to accepted procedures, the data will not only be reliable, it will also be less costly.
3. Accessibility to all elements within the universe: in a very large population such as students of Nigerian polytechnics or practicing farmers in Nigerian. It is difficult, costly and time consuming to locate the entire students or farmers.

Types of samples

The two main categories of sample are:

1. Random samples (or probability samples)

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If $\chi^2_{\text{cal}} > \chi^2_{\text{tab}}$ mean reject H_0 and accept H_a
 If $\chi^2_{\text{cal}} < \chi^2_{\text{tab}}$ mean accept H_0 and reject H_a

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Non-random sample (or non-probability samples).

Non-random sample are easier and less costly to obtain. It should be noted that researcher's sample should be representative of the particular population or universe from which it is drawn. If the population is homogenous, sampling will be much easier and any sample is likely to be representative. But if the population is heterogeneous, great care must be taken in selecting a sample of heterogeneous population, which should be larger than those of homogenous population.

Probability sampling

There are four sampling techniques under probability sampling;

- 1- Random sampling
- 2- Systematic sampling
- 3- Stratified random sampling;
- 4- Cluster sampling (or are sampling)

1.

Random sampling: this is best known type of probability sample. Probability of being chosen for the sample and every collection of persons of the same size has an equal probability of becoming the actual sample. This probability is n/N , where n stands for the size of the sample unit and N the size of the population.

Advantage of random sampling: randomness pit biases and provides statistical means for estimating sampling errors.

2. Systematic sampling techniques

This consists of selecting every k^{th} sampling unit of the population after the first sampling unit has been selected at random from the total of sampling unit. For example, if you wish to select a sample of 100 persons from a population of 10,000 you will take every 100th individuals ($k=100$). The first selection will be based on some random process such as the use of a table of random digits. Suppose that the 14th person were selected, the sample would then consist of individual's numbers 14, 114, 213 and so on.

Advantages: systematic sampling is simpler to random sampling. It also more practical when one does not have a sampling frame completed in advance but must compile the frame during the course of the sampling or interviewing.

Stratified random sampling

This is another form of random sampling separating the population elements into non-overlapping groups called strata and then, selecting a simple random from each stratum to obtain a stratified sample. Example, if a researcher is conducting a study among polytechnic lecturers in Federal Polytechnic Mubi, he does not need to get in touch with all lecturers from Assistants lecturers, to chief lecturer. Since this group is rank-ordered, he can stratify them based on their status, chief lecturers, principal lecturers, senior lecturers, lecturer I, lecturers II, and Assistant Lecturers. That is, he should list all chief lecturers together in one homogenous group, then to all other groups. After this, he draws a systematic sample from each group.

Assuming it was required to sample 2000 students of the school of Agriculture who's their entire population be stratified as follows.

S/No	Student's Homogenous groups	Group's Stratification
1.	HND-II	200
2.	HND-I	400
3.	ND-II	600
4.	ND-I	800
	Total	2000 Students of the sch. Of Agric.

Sample Calculations

S/No	Groups	Computations	Group's sample
1.	HND-II	$\frac{200 \times 100}{2000}$	10
2.	HND-I	$\frac{400 \times 100}{2000}$	20
3.	ND-II	$\frac{600 \times 100}{2000}$	30
4.	ND-I	$\frac{800 \times 100}{2000}$	40
	Total		100 sampled students

Advantages: This method avoids the possible bias of taking a systematic sampling from a non-stratified sampling frame and it also saves time and money. A stratified sample equally assures representation not only of all ranks but simultaneously of all age and income within the population being sampled.

Cluster sampling

It is frequently used in large scale studies because it is the least expensive sample design. Cluster sampling involves first selecting large groupings called clusters and then selecting the sampling units from the cluster. The clusters are selected by a simple random

sampling or a stratified sample. Depending on the research problem, researchers can include all the sampling units in these clusters or make a selection from within the clusters using simple or stratified sampling procedure. These sampling techniques are applicable to any type of design depending on the direction of the research.

Advantage: the main advantage of cluster sampling is that it saves time and money. It obvious advantage is that it is not a single sample but two or more within a possibility for sampling error each case; non-probability sampling.

2 Non-probability sampling

There are six basic types of non-probability sampling

- 1- Quota sampling
- 2- Multistage sampling
- 3- Panel sampling
- 4- Purposive or judgmental sampling
- 5- Snowball sampling
- 6- Convenience or accidental sampling

Non-probability sampling has one obvious advantage. It is less complicated, much less expensive to take advantage of available respondents (example volunteers) without the statistical complexity of a probability sample.

Non probability sampling has some **disadvantages**. Since the probability that a person will be chosen is not known. The researcher generally cannot claim that his/her sample is representative of the large population or not. This therefore limits the researcher's ability to generalize his/her finding beyond the specific sampled studied.

1. *Quota sampling:* a quota sample is the non-random equivalent of a stratified random sample with the added requirement that each stratum is generally represented in the sample in the same proportion as in the entire population. Quota is sub divided into various categories example male/female, old/young, working/unemployed and soon. The method is cheap and reasonably effective and also widely used.

2. *Multistage sampling:* This is a practical system widely used to reduce the traveling time for interviewers and the subsequent cost. Its similar to stratified sampling except that the groups and sub-groups are selected on a geographical/location basis rather than some social characteristic. The idea here is large cluster of population are divided in to smaller clusters in several stages in order to make primary data collection more manageable.

Application of multistage sampling steps

- i- chose sampling frame
- ii- chose sampling frame of relevant sub-groups from previous stage
- iii- repeat the second stage above if necessary.
- iv- chose the members of the group from the sub-groups.

Illustration Topic: Evaluate online spending patterns of households in Nigeria.

You can sample a group of 120 household as follows:-

- choose 6 states from the six geopolitical zones using simple random sampling.
- choose 4 local governments out of the six selected state using systematic sampling method.

-choose 5 households from each of the local govts. Sampled using simple random sampling method ($6 \times 4 \times 5 = 120$ households).

3. *Panel sampling:* This is a method of selecting a group of participants through a random sampling method and then asks that group for the same information again several times over a period of time. This allows for estimates of changes in the population.

4. *Convenience sampling (or accidental sampling):* in convenience or accidental sampling, the researcher simply chooses the closest live person as respondents, what is lost in sampling accuracy is saved/gained in time and money.

5. *Purposive sampling (or judgmental sampling)* this does not involve a quota to fill from within various strata as in quota sampling. Rather, the researcher uses his/her own judgment about which respondents to choose and picks, only these who best meet the purpose of the study.

The obvious advantage of purpose sampling is that the researcher can use his/her research skill and prior knowledge to choose respondents.

Snowball sampling: researchers conducting observational research and community studies often use this method. It is usually graduated in stage. In stage one, a few person having the requisite characteristics are identified and interviewed. These people are used as informants to identify others who qualify as inclusion in the sample. The second stage involves interviewing these persons who in turn lead to still more persons who can be interviewed in the third stage and so on. The term snow ball stems from the analogy of a snowball which begins small but becomes bigger and bigger as it rolls downhill.

Sampling size

Most introductory researcher courses are not explicit on a minimally adequate sample size. It should be noted however, that the correct or adequate sample size is contingent on the nature of the population and the purpose of the study. For studies dealing with small population for which the entire population may number ten (10) or less, the entire population should be sampled.

Questionnaire

A questionnaire is simply a series of questions developed by a researcher to be answered by the respondents in order to investigate/study a phenomenon. There are two ways of collecting information/data, namely by questionnaires and by interviews. Essentially you gather data/information by asking people for the data/information you want and by observing respondent's behavior or the products of their behaviours.

General importance of questionnaire;

1. It is the most widely used instrument for data collection in educational researcher.
2. They provide a guide means of obtaining respondent's view on a wide range of subjects.
3. In direct-contacts and in interviewing research subject which is expensive and time Consuming, demands are substantially reduce through the uses of questionnaire.
4. Again, the questionnaire is more efficient and practical compared to interview, because of personal appearance, mode, conduct and interview skills of the investigator.

Questionnaires are usually of two forms or types, close or structured type and open or unstructured type. A closed or restricted type, questionnaire restricts the respondent to the range of possible answers provides. Consider this example of closed or restricted types, which you may observe is short clear and straight forward.

1. Why did you choose to do your practical at the polytechnic farm
 - a. Because of convenience of location
 - b. Because of better research facilities
 - c. Because of the polytechnics reputations
 - d. Because of no space selected elsewhere
 - e. Because of lower mobility cost

An open or unstructured questionnaire gives more leeway to the respondent to freely respond to an item in his own words, consider this example;

1. Please indicate below why you choose to do your practical at the polytechnic farm.....

Some questionnaires combine closed and open items depending on what the researcher wants to know from the respondent. Each type has its strengths and weaknesses. For the questionnaire itself, where open or closed or both, its preparation would influence its quality.

Rating scale: this is an influence used for increasing personality, reliable and geared toward attaining summary information for investigating ones research problem.

Most usually a numerical value is attached to the categories and the numerical value usually reuses from 1 to 7, with 1 being the lowest value and 7 being the highest value. The types of rating are retrieval to as numerical rating scale. An example is shown below.

Q=What are the problems you encountered in your Cattle's marketing, rate them in ascending order from 1 – 7?

PROBLEMS*	1	2	3	4	5	6	7
Inadequate Capital							
High cost of feed and water							
Inadequate medical services							
Theft and Rubbery							
Mortality on transit							
High taxation							
Availability of Cattle							

We also have a **Graphical rating scale** in this type; the researcher is expected to place a check on a vertical line and the subject at the horizontal line. Example student ages in a class as their check and their registration number on the horizontal line.

Guideline (recommendations) for preparing questionnaire

- 1- Link a set of items sequentially under one message or theme, of which these items should be derived from one research question.
 - 2- Ideally three to five items should serve the basis of answering one research question from which the items are derived.
 - 3- Keep the total questionnaire as brief and clear as possible, so that it is very readily understood and can be completely filled in with ease and over short time.
 - 4- Use construct and words that respondents are likely to be familiar with and have common agreement about their meaning.
 - 5- Avoided in complete alternatives: married? Yes - No
 - 6- Arrange question in neat, clear, orderly sequence in such a way that responses can be analyzed easily.
 - 7- A brief letter of introduction about the aims of the study and confidentiality of responses should accompany questionnaire sent to the respondents. This enables the researcher to explain the aims of the questionnaire and persuade the subject to respond to it.
 - 8- When response is as low as 40-50%, the data obtained from such a low return rate are of limited and doubtful usefulness given that it is possible that if more people responded, say an additional 40%, the information would have severely altered the results.
 - 9- Avoid questions which embarrass and put respondents on the defensive. For instance, a questionnaire should not ask respondents whether they have degree or school certificate, married or bachelor but rather a question can be used to ask them the highest grade they completed before leaving school.
- Nevertheless, questionnaires are very useful and appropriate instruments. For gathering data in educational research, hopefully the recommendations would enable one develop a valid, reliable and appropriate questionnaire for use in research data collection.

Interview studies

The research interview has been defined as two person conversation initiated by the interviewer, for the specific purpose of obtaining researcher relevant information/data, and focused by him/her on content specified by research objectives, system description, prediction or explanation. Interviews involve human interaction. Interview may vary from a brief, structured session to a lengthy complicated unstructured session lasting some hours.

The structured interview uses a schedule which is mainly in questionnaire that is read to respondents in a specified order.

The structured interview is easy to score; reduces interview bias, is more easily replicated and is more reliable than an unstructured interview, is that the interviewer can explore the respondent's/interviewers answer. It is unstructured because there is no question written down in advance with the researcher assigning only a topic, he composes the questions as it goes a long.

Note: the procedures used for constructing questionnaires are essentially the same as those used for constructing interview schedules. An interview may and may not be accompanied by a cover or introduction letter.

Advantages of interview studies

1. *Flexibility*: specific questions can be probed and a question can be repeated when the response indicate that the respondent/interviewee misunderstood.
2. *Response rate*: it has a better response rate (than mails questionnaire) because many people feel more confident over their speaking ability than their writing ability.
3. *Non-verbal behaviours*: this can be observed to assess the validity of the respondent answer.
4. *Control over environment*: the environment can be standardized by making sure that the interview is conducted in privacy, convenient that there is no noise among other things.
5. *Question order*: it ensures that the respondent does not answer the question out of order.
6. *Completeness*: it ensures that all the questions are answered.
7. *Time of interview*: time, data and place of interview can be recorded.
8. *Greater complexity of questionnaire*: a complex question can be used in an interview study.
9. *Respondent alone can answer*: this prevents "aided" answer, i.e. it prevent respondents from receiving information from other persons.

Disadvantages of interview studies

- 1- *Cost*: it can be extremely costly e.g. travel expenses.
- 2- *Time*: interviews are lengthy and may require the interviewer to travel far, interviewer must arrange the time for when the respondent is at home.
- 3- *Interview bias*: the respondent's answers can be affected by his/her reaction to the interviewer's sex, race, social class, age, dress and physical appearances.

4. No opportunity to consult records.

5. *Spontaneity*: a person's reasoning ability is adversely affected by such factors as fatigue, stress, illness, heat and density.

6. *Less anonymity*: the interviewer typically knows the respondent, name and address.

Note: some of the advantage and disadvantage are debatable.

Approaching the respondent/interviewers

The interviewer on approaching the interviewee performs all functions that the cover letter performs. The introduction tasks of the interviewer include.

1. Tell the respondent who he/she is and who he/she represents, it possible showing an ID card.
 2. Tells the respondent what you are doing such as what the study is all about and assure him/her of enormity/confidentiality.
 3. Tell the respondent why he/she was chosen emphasizing that he/she was not singled personally for harassment but was chosen in an impersonal way merely because a cross section of the population is needed.
 4. Use a positive approach: for instance, the respondent may be busy for the interview, approach him/her by saying, I would like to come in and talk with you about this "rather than, may I come in or should I come back later?"
- Note: it is good to be polite and courteous when approaching the interviewee. Dress appropriately and be friendly.

Procedure for conducting the interview

There are three main procedures to follow when conducting the interview;

1. Ask question as worded: uses standardized questionnaire with every respondent asked the same questions, mix clarification of questions should be avoided as much as possible, since it can lead to subtle changes in question meaning, but questions can be repeated when the respond request for it.
2. Ask question in order: attention is usually given to a questions order when writing the questionnaire. The question order should be retained, if the respondent refuses to answer a question, this best procedure is to merely record the lack of response and proceed to the next question.
3. Do not lead respondents: reading the questions as stated will help the interviewer guard against biasing or leading, even when some respondents feel a need to know the interviewer's position on a particular question, it is important to make sure that the respondent does not feel as though he/she taking an examination or on trial.

Note: always come to the level of the interviewers, functionality is important be time conscious be precise and concise.

RESEARCH REPORTING

All research reporting use roughly the same format. It doesn't matter whether you have done a survey research or experimental research all have the same basic structure and format. the rationale is that readers of research reports (i.e. decision makers, funders, students, policy makes etc) will know exactly where to find the information they are looking for, regardless of the individual report. Once you have learned the basic rules for research proposal and report writing, you can apply them to any research discipline. The same rules apply to writing a proposal, a thesis, a dissertation, or any business research report. Research paper usually has five chapters with well-established sections in each chapter. Readers of the paper will be looking for these chapters and sections, so you should not deviate from the standard format unless you are specifically requested to do so by the research sponsor.

Most research studies begin with a writing proposal. Again, nearly all proposals follow the same format. In fact, the proposal is identical to the first three chapters of the final paper, except that its writing in future tense. In proposal, you might say something like "the researcher will secure the sample from" while in the final paper, it would be changed to "the researcher secured the sample from"

The most common used style for writing research report is called "APA". And the rules are described in the publication manual of the American Psychological Association (APA). You can find it readily available, in many libraries. The style guide contains hundreds of rules for grammar, page size and layout, font size, abstracting, citation of references etc.

Such rules avoid the use of first person pronouns. Refer yourself or the research team in third person instead of saying "I will or we will" just say something like "the researcher will" or "the research team will" Also instruction like "paper shall be white and of size A4" or "Thesis must be typed double spaced using Time new Roman", "the page number shall be type written at the center of the Top of each page", "Abstract shall be between 200 to 250 words and shall contain the followings:-

- i. A brief statement of the specific objectives and shows contributions of the research work.
- ii. A brief statement of research methodology used including methods of data analysis.
- iii. A brief straight to the point summary of specific findings

iv. A brief statement of implication of the findings(Conclusion).

Chapter one shall contain the following:-

- i. A general overview of the background concept and general or broad objective of the research.
- ii. Specific objectives of the research work or working Hypothesis as the case may be.
- iii. Scope and limitations of the research.

Chapter two (Literature review):- This chapter deals with incisive, coherent and up to date review of relevant literature in research in research area. Student or researcher expected to show a high proficiency in information retrieval and interpretation. The literature review shall geared toward justifying the define objectives of the research.

Chapter three(Materials and Methods):- This chapter shall contain a list of specialized materials and a detailed description of scope/Area of the study and methods employed in the research work. E.g. Tools of analysis, chemical, reagents, living organism, material, apparatuses/gadgets and equipment that are used in the laboratory.

Chapter four(Data analysis/Results or findings and Discussions) as follows:-

- i. Results/findings-Here the outcome of the research shall be presented and explain in this section. The findings shall be in prose and references made to tables, figures or plate in the texts.
- ii. Discussion- this section shall devoted to giving a comprehensive account of the findings of the research and relating them to published works. Special attention should be given to significant or novel findings

Chapter five (Summary, Conclusion and Recommendation):- This chapter shall highlight the major findings of the research work and draw conclusion from the analysis and make recommendations.

*References:-*References be shall cited in the text by means of the authors surname, follow by the year of publication. "If the authors name form part of a sentence in the text, the year of publication shall be given in parenthesis. E.g. it was discovered that Ahmed(2003) noted the difference in price. "If authors name does not form part of a sentence in the text, both the authors name and year of publication shall be given in parenthesis at the end of the sentence or after an appropriate part. e.g. "in a recent study (Ahmed, 2003) it was discovered that..." A work of two or more authors but less than 5 authors the names of the two authors shall be cited followed by the year of publication i.e. 1st and 2nd authors e.g. "In a related study of assessment for promotion exercise(Bolaji and Emmanuel, 1998) it was found

Table of Chi-square statistics

P-values: P=0.05 | P=0.01 | P=0.001

df P = 0.05 P = 0.01 P = 0.001

1	3.84	6.64	10.83
2	5.99	9.21	13.82
3	7.82	11.35	16.27
4	9.49	13.28	18.47
5	11.07	15.09	20.52
6	12.59	16.81	22.46
7	14.07	18.48	24.32
8	15.51	20.09	26.13
9	16.92	21.67	27.88
10	18.31	23.21	29.59
11	19.68	24.73	31.26
12	21.03	26.22	32.91
13	22.36	27.69	34.53
14	23.69	29.14	36.12
15	25.00	30.58	37.70
16	26.30	32.00	39.25
17	27.59	33.41	40.79

$$\frac{15}{12} = \frac{14}{11}$$

$$\frac{11}{11} = \frac{11}{11}$$

$$\frac{27}{17} = \frac{17}{17}$$

$$\frac{12}{12} = \frac{12}{12}$$

$$\frac{34}{42} = \frac{34}{42}$$